

Dr. Sarah Bürli

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Employment

01.2025 – present	Head of the Botanical Garden of the University of Graz, Austria (AT).
05.2023 – 11.2024	Postdoctoral researcher, Spatial Ecology Group of A/Prof Bradley Case and Prof. Hannah Buckley, Auckland University of Technology (AUT), New Zealand (NZ).
10.2022 – 03.2023	Postdoctoral researcher, Ecology Group of Prof. Sabine Rumpf, University of Basel, Basel, Switzerland (CH).

Education

2018 – 02.09.2022	PhD thesis ès Science in Biology “Causes & consequences of the rarity of plant species”, Botanic Garden of the University of Bern, Bern, CH - Advisors: Prof. Markus Fischer & Dr. Andreas Ensslin.
2017 - 2018	Master project “Environmental effects on sex ratio and sexual leaf dimorphism variation in dioecious plants”, University of Lausanne, Lausanne, CH - Advisor: Prof. John Pannell.
2016 - 2018	Master ès Science in Biology, section “Behaviour, Evolution and Conservation”, University of Lausanne, Lausanne, CH.
2016	Project “Temperatures and growing season length define the upper limits of grasslands in European mountain”, University of Lausanne, Lausanne, CH - Advisor: Dr. Pascal Vittoz.
2013-2016	Bachelor ès Science in Biology, University of Lausanne, Lausanne, CH.
2011-2013	Study in Physics, EPFL, Lausanne, CH.

Training & workshop participation

October 2025	Leadership Lab for Researchers I & II: Expanding your toolbox of leadership skills, Dr. Yasmin Dolak-Struss, Uni Graz, Graz, AT.
June 2025	Individual coaching sessions in people, resource and time management, and leadership, Dr. Barbara Schneeberger, Graz, AT.
January 2025	<i>Ex-situ</i> conservation of threatened plant species Workshop III, Botanic Garden of the University of Bern, Bern, CH.
May 2024	“Bayesian Species Distribution Models” workshop, PR Statistics, Dr. Marcia Barbosa, online. “Coaching essentials” workshop, AUT, Auckland, NZ. “Bystander intervention” workshop, AUT, Auckland, NZ.
November 2023	New Zealand’s first aid certificate, Hato Hone St John, Auckland, NZ.
July 2023	Python coding workshop, AUT, Auckland, NZ.
February 2022	“Performing in the storm” workshop, University Conference of Western Switzerland (CUSO), Dr. T. Teichler, online.
November 2021	“Introduction to Research Data Management” workshop, University Library Bern, online.
October 2021	“Project Management” workshop, Dr. S. Henrichs, University of Bern, Bern, CH.
May 2021	“Effective Grant Writing for Young Scientists” workshop, CUSO, Prof. M. Toscano, online.
March 2021	“Plant Translocation” workshop, Prof. T. Abelli, online.
January 2021	“Proactively Managing Relationships and Networks” workshop, CUSO, Dr. T. Wranik, online. <i>Ex-situ</i> conservation of threatened plant species Workshop II, online.
April 2020	“Effective doctoral research, use, and management of scientific publications” workshop, CUSO, T. Henkel & L. Mellifluo, online.
March 2020	Volunteer at the World Biodiversity Forum, Davos, CH.

February 2020	"Negotiation Skills" workshop, CUSO, Dr. F. Salina, Geneva, CH.
November 2019	"Time and priority management" workshop, CUSO, S. d'Autherville, Geneva, CH.
September 2019	"The Temporal Dynamics of Evolution" workshop, CUSO, Prof. T. Flatt, Geneva, CH.
July 2019	Swiss Field Botany Certificate 400 species, Info Flora, Bern, CH.
April 2019	"Structural Equation Modelling" workshop, CUSO, Dr. J. Grace, Fribourg, CH. Ex-situ conservation of threatened plant species Workshop I, Botanic Garden of the University of Bern, Bern, CH.
July 2018	Swiss Field Botany Certificate 200 species, Info Flora, Bern, CH.
July 2017	International Summer School on alpine plant life, Prof. C. Körner, Furka Pass, CH.
June 2017	Evolutionary Biology Workshops in the Alps, Prof. T. Kawecki, Aletsch, CH.
Summers 2016 & 2015	Internships at the Botanic Gardens and Museum of the Vaud canton, Lausanne, CH.

Workshop and conference organisation

November 2026	(Planned) Co-organisation of the conference entitled "A relational World?", with Dr. Caroline Gatt & Prof. Nassim Balestrini, University of Graz, Graz, AT.
March 2022	Co-organisation of the workshop entitled "Towards a more inclusive science: Women and genderqueer people in academy", invited speaker: Dr. Sabine Höfler, University of Bern, Bern, CH.

Presentations & media interviews

February 2026	(Planned) "Reparative ecology: how to restore native forests in Aotearoa New Zealand" talk, Vaud Canton's Botany Circle (invited speaker), Lausanne, CH.
January 2026	(Planned) "Restoring Forests Through Partnership: Lessons from Work with Māori Communities" talk, Biological Colloquium, University of Graz, Graz, AT.
November 2025	Invited speaker at the panel talk "30% Dandelion or the Blooming relationships in the vegetal turn: art, science & politics", Graz, AT.
September 2025	Interviewed by the journals Gärtner+Florist and LAK (Steiermärkische Landarbeiterkammer), Graz, AT. Interviews not available online.
August 2025	Interviewed by the journal Die Krone, Graz, AT. Interview available at https://www.krone.at/3938828
May 2025	Interviewed by the journal Die Presse, Graz, AT. Interview available at https://www.diepresse.com/19640785/die-neue-leiterin-sarah-buerli-hat-die-maori-als-vorbild .
March 2025	Interviewed by the University of Graz's communication team, Graz, AT. Interview available at https://www.uni-graz.at/en/news/mensch-und-natur-sarah-buerli-ist-neue-wissenschaftliche-leiterin-des-botanischen-gartens/
February 2025	Interviewed by the journal Die Kleine Zeitung, Graz, AT. Article available at https://www.msn.com/de-at/lifestyle/other/natur-im-umbruch-es-gr%C3%BCnt-so-fr%C3%BCh/ar-AA1z8rX8
April 2024	Interviewed by New Zealand's Biological Heritage National Challenges, Auckland, NZ. Interview available at https://bioheritage.nz/a-postdoc-for-aut-living-labs/ . "How to restore native forest and human relationships in Aotearoa New Zealand?" talk, New Zealand's Biological Heritage National Challenges conference (invited speaker), Wellington, NZ.
September 2023	"Living Laboratories Research Planning Workshop" talk, AUT, Auckland, NZ.
October 2022	"What makes a plant species rare and how can we improve its conservation?" talk, Plant and

	animal conservation seminar (invited speaker), University of Bern, Bern, CH.
September 2022	"Drivers of plant rarity and consequences for their conservation" talk, Research group of Prof. A. Phillimore, University of Edinburgh, Edinburgh, UK.
April 2022	"How genetic diversity, propagule pressure and ecological distance interact to affect early reintroduction success in rare and threatened plant species?" talk, Ecology Seminar, University of Bern, Bern, CH.
November 2021	"Do rare plants suffer more from inbreeding and outbreeding depression than common plants?" talk, Ecology Seminar, University of Bern, Bern, CH.
June 2021	"Are rare plants less well defended than common ones?" talk, Biology Day of the University of Bern, Bern, CH.
December 2020	"Does herbivory drive plant rarity? New insights" talk, British Ecology Society Conference, online. "What does herbivory tell us about plant rarity" talk, Ecology Seminar, University of Bern, Bern, CH.
November 2020	"Ex situ conservation and reintroduction of endangered flowering plants – 2018-2020" talk, 2nd meeting with the Swiss Federal Office for the Environment, University of Bern, Bern, CH.
December 2019	"Effect of genetic diversity and propagule pressure on the reintroduction success of rare plant species of Switzerland" talk, Ecology Seminar, University of Bern, Bern, CH.
June 2019	"Ex situ conservation and reintroduction of endangered flowering plants" talk, Kick-off meeting with the Swiss Federal Office for the Environment, University of Bern, Bern, CH.
May 2019	"Does herbivory drive plant rarity? An experimental multi-species approach" talk, Pop Bio conference, Warsaw, Poland.
November 2018	"Causes and consequences of plant rarity" talk, Ecology Seminar, Institute of Plant Sciences, Bern, CH.
February 2018	"Temperature threshold for grasslands in alpine-nival ecotone" talk, Alpine conference, Fribourg, CH.

Teaching

2025 - present	Teaching of the following courses at Uni Graz, Graz, AT: Undergraduate courses: <i>Nature & Species Conservation</i> and <i>Plants of Austrian native habitats</i> . Graduate courses: <i>Integrative Taxonomy</i> , <i>Plant Conservation</i> and <i>Vegetation Distribution & Climate Change</i> .
2024	Teaching assistant for the undergraduate course <i>Botany and Taxonomy</i> , AUT, Auckland, NZ.
2017 - 2022	Teaching assistant for several undergraduate courses, including practical ones, Universities of Lausanne and Bern, Lausanne and Bern, CH.
2006 - 2016	Learning tutor at the Vaud Youth Assistance Center, Lausanne, CH.

Mentoring

2025 – present	Supervision of K. Mayr's research project " <i>Reintroduction of the endangered Adenophora liliifolia</i> ", B. Papavassilis' research project " <i>Effects of climatic variation on the rare translocated plant species</i> " and M. Klug's Master thesis " <i>Impact of land-use intensity on orchid diversity in meadows around Graz</i> ". Co-supervision of B. Ranner's Master thesis " <i>Impact of Climate and Land Use on Micro-rodent Diversity</i> " and S. Wolf's Master's thesis " <i>A Century of Forest Vegetation Dynamics Around Graz: Quantifying Long-term Changes and Evaluating the Role of Forest Management on Phytodiversity</i> " with Assoc. Prof. A. Clark. Co-supervision of S. Döhrn's Bachelor thesis " <i>Decolonisation methods of a Botanical Garden collections</i> " with Dr. M. Wurzer.
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2020 - 2021	Co-supervision of A. Abanto's Master thesis "Effects of fertilization and water stress on the growth, pollen and seed fitness of closely related rare and common plant species of Switzerland", University of Bern, Bern, CH.
2019	Co-supervision of C. Senn's Bachelor thesis "Caterpillar, snail and slug bioassays with rare and common plant species", University of Bern, Bern, CH.

Fellowships & grants (PI or co-PI)

November 2025	Research Culture and Integrity Ambassador Grant 2025 from the Research Career Campus Uni Graz, project title: "From roots to words: Rethinking How We Speak About Plants (and Us)". https://research-careers-campus.uni-graz.at/de/neuigkeiten/who-are-the-uni-graz-research-culture-and-integrity-ambassadors/
May 2025	Arqus Innovation Fund, ad hoc bottom-up initiative, project title: "Arqus Verde: Connecting Botanical Gardens for Europe's Biodiversity and Cultural Heritage". https://arqus-alliance.eu/arqus-innovation-fund/
March 2025	Oak Spring Garden Foundation's Plant Ecology Interdisciplinary Residency Fellowship 2025, Virginia, USA. https://www.osgf.org/blog/2025-interdisciplinary-residents-announcement
March 2021	Travel Grant from Cost Action & Conserve Plants (CA18201).

Publications

Peer-reviewed publications:

Bürli, S., McClean, L. A., Case, B. S., David Hall, D., & Buckley, H. L. (in review in *Ecological Solutions and Evidence*). Seedling survival of late-successional tree species: Results from three forest restoration experiments.

Bürli, S., Buckley, H. L., & Case, B. S. (in review in *Ecology and Evolution*). Effects of biotic and abiotic modifiers and experimental treatments on tree seedling recruitment: A systematic review.

Bürli, S., Fischer, M., & Ensslin, A. (2025). Disentangling the effects of population mixing and propagule amount in rare plant translocations. *Biological Conservation*, 306, 111132. DOI: 10.1016/j.biocon.2025.111132

Godefroid, S., Lacquaye, S., Ensslin, A., Dalrymple, S., Abeli, T., Branwood, H., Ferrando Pardo, I., Ferrer Gallego, P.P., Zippel, E., Gouveia, L., Lobo, C.A., Fernandes, F., Rasp, M., Daco, L., Séleck, M., Frankard, P., Fischer, L.K., Koutsovoulou, K., Finger, A., Valkó, O., Garfi, G., Španiček, B., **Bürli, S.**, Dickenmann, R., Jones, J., Guyonneau, J., Papugua, G., De Vitis, M., Fenu, G., Van Rossum, F., Cogoni, D., Foster, G., Julien, M., Piqueray, J., Vít, P., Bonnet, V., Carta, A., Descombes, P., Lazarević, M., Müller, N., Orsenigo, S., Ravera, S., Sild, M., Smyth, S., Wagner, T.C., Walisch, T., Ballian, D., Cambria, V.E., Colas, B., Dixon, L., Draper, D., Froidevaux, V., Pagitz, K., Pánková, H., Pankhurst, T., Preston, J., Prunier, P., Saura-Mas, S., Sartori, K., Socher, S.A., Stanik, N., Wunderli, A., Nikowitz, T., White, F.J. (2025). Current state of plant conservation translocations across Europe: Motivations, challenges and outcomes. *Biodiversity and conservation*, 1-24. DOI: 10.1007/s10531-025-03013-0 | <https://rdcu.be/d692m>

Stewart, G. T., Perrott, J., Buckley, H. L., **Burli, S.**, Keiha, P., Walker, L., Henare, D., & Wilkie, K. (2024). Commentary: Respectful relations between science and Māori knowledge. *ACCESS: Contemporary Issues in Education*, 44(1). DOI: 10.46786/ac24.5235

Bürli, S., Ensslin, A., & Fischer, M. (2024). Rarity and life-history strategies shape inbreeding and outbreeding effects on early plant fitness. *Global Ecology and Conservation*, e03081. DOI: 10.1016/j.gecco.2024.e03081

Bürli, S., Ensslin, A., Kempel, A., & Fischer, M. (2023). Are rare plant species less resistant than common ones to herbivores? A multi-plant species study using above- and below-ground generalist herbivores. *Ecology and evolution*, 13(9), e10482. DOI: 10.1002/ece3.10482

Bürli, S., Pannell, J.R., & Tonnabel, J. (2022). Environmental variation in sex ratios and sexual dimorphism in three wind-pollinated dioecious plant species. *Oikos*, e08651. DOI: 10.1111/oik.08651

Bürli, S., Theurillat, J., Winkler, M., Lamprecht A., Pauli, H., Rixen, C., Steinbauer, K., Wipf, S. Abdaladze, O., Andrews, C., Barančok, P., Benito-Alonso, J., Fernández Calzado, M., Carranza, M., Dick, J., Erschbamer, B., Ghosn, D., Gigauri, K., Kazakis, G., Mallaun, M., Michelsen, O., Moiseev, D., Moiseev, P., Molau, U., Molero Mesa, J., Morra di Cella, U.,

Nadeem I., Nagy, L., Nicklas, L., Palaj, A., Pedersen, B., Puşcaş, M., Rossi, G., Stanisci, A., Tomaselli, M., Unterluggauer, P., Ursu, T., Villar, L., & Vittoz, P. (2021). A common soil temperature threshold for the upper limit of alpine grasslands in European mountains. *Alpine Botany*, 131(1), 41-52. DOI: 10.1007/s00035-021-00250-1

Other publications:

Wilfling, J., Brosch, U., Berg, C., & **Bürli, S.** (*in press*). Development of the historical greenhouses of the University of Graz since 2021.

Case, B. S., Buckley, H. L., & **Bürli, S.** (2024). Tackling two crises with one Nature-based Solution. PureAdvantage. <https://pureadvantage.org/tackling-two-crises-with-one-solution/>

Bürli, S. (2022). *Causes and consequences of the rarity of plant species*. (Thesis). Universität Bern, Bern. DOI: 10.48549/3754. <https://boristheses.unibe.ch/3754/>

Scientific reviewing activities

Reviewer for the journals *Oikos*, *Annals of Botany* and *Restoration Ecology*.

Professional services

2025 – present	Biodiversity buddy (i.e. being a contact point for questions and support in challenging situations related to diversity) at the University Graz, Graz, AT.
	Expertise cluster “Climate Change Graz”, University of Graz, Graz, AT.
	Expertise cluster “Biodiversität”, University of Graz, Graz, AT.

Membership and committee

2025 – present	Friends of the Botanical Garden of the University of Graz, Graz, AT.
	Malvine Foundation, Graz, AT.
2020 – present	Swiss Botanical Society, CH.
2017 - 2018	Student representative for a Professor recruitment at the Faculty of Medicine and Biology, University of Lausanne, Lausanne, CH.
2016 – present	Member of the committee of Liber & Labor association, Lausanne, CH.

Languages

French	Mother tongue (C2)
English	Excellent (C1)
German	Good (B2)
Italian	Basic (A2)

IT

R, R Markdown, Python, C++, Git, GIS, Office & Libre Office suites